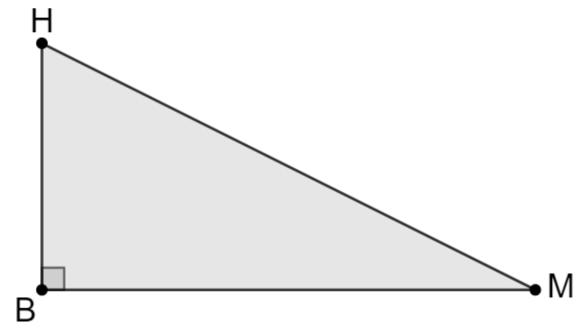
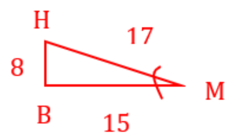
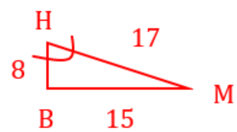
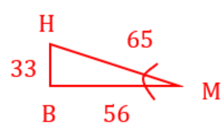
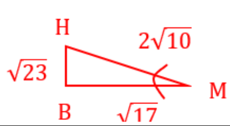
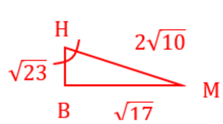
Geometry
Unit 7
Lesson 3 Homework

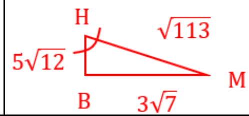
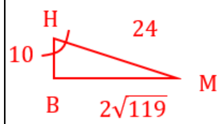
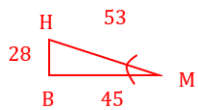
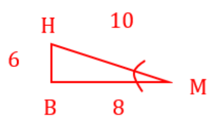
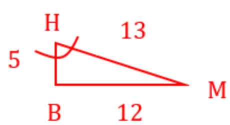
Name_Answer Key__________
Date______________________
Period_____________________

Triangle HBM is shown where $m\angle B = 90°$.



Complete the table using the given information.

	HB	BM	HM	Reference Angle	Cosine	Sine	Tangent
1.	8	15	17	$\angle M$ 	$\frac{A}{H} = \frac{15}{17}$	$\frac{O}{H} = \frac{8}{17}$	$\frac{O}{A} = \frac{8}{15}$
2.	8	15	17	$\angle H$ 	$\frac{A}{H} = \frac{8}{17}$	$\frac{O}{H} = \frac{15}{17}$	$\frac{O}{A} = \frac{15}{8}$
3.	33	56	65	$\angle M$ 	$\frac{A}{H} = \frac{56}{65}$	$\frac{O}{H} = \frac{33}{65}$	$\frac{O}{A} = \frac{33}{56}$
4.	$\sqrt{23}$	$\sqrt{17}$	$2\sqrt{10}$	$\angle M$ 	$\frac{A}{H} = \frac{\sqrt{17}}{2\sqrt{10}}$	$\frac{O}{H} = \frac{\sqrt{23}}{2\sqrt{10}}$	$\frac{O}{A} = \frac{\sqrt{23}}{\sqrt{17}}$
5.	$\sqrt{23}$	$\sqrt{17}$	$2\sqrt{10}$	$\angle H$ 	$\frac{A}{H} = \frac{\sqrt{23}}{2\sqrt{10}}$	$\frac{O}{H} = \frac{\sqrt{17}}{2\sqrt{10}}$	$\frac{O}{A} = \frac{\sqrt{17}}{\sqrt{23}}$

6.	$5\sqrt{2}$	$3\sqrt{7}$	$\sqrt{113}$	$\angle H$ 	$\frac{A}{H} = \frac{5\sqrt{2}}{\sqrt{113}}$	$\frac{O}{H} = \frac{3\sqrt{7}}{\sqrt{113}}$	$\frac{O}{A} = \frac{3\sqrt{7}}{5\sqrt{2}}$
7.	10	$2\sqrt{119}$	24	$\angle H$ 	$\frac{A}{H} = \frac{10}{24}$	$\frac{O}{H} = \frac{2\sqrt{119}}{24}$	$\frac{O}{A} = \frac{2\sqrt{119}}{10}$
8.	28	45	53	$\angle M$ 	$\frac{A}{H} = \frac{45}{53}$	$\frac{O}{H} = \frac{28}{53}$	$\frac{O}{A} = \frac{28}{45}$
9.	6	8	10	$\angle M$ 	$\frac{A}{H} = \frac{8}{10}$	$\frac{O}{H} = \frac{6}{10}$	$\frac{O}{A} = \frac{6}{8}$
10.	5	12	13	$\angle H$ 	$\frac{A}{H} = \frac{5}{13}$	$\frac{O}{H} = \frac{12}{13}$	$\frac{O}{A} = \frac{12}{5}$